HDOC DAY 61

PROBLEM

Write a C program that reads the size and elements of a variable-length integer array, nums[], containing only positive integers, and an integer target. Pass the array to a function named two_sum() using its address. Use pointers to pass the array between the calling function and two_sum(). The function two_sum() must:

- Find two distinct indices, i and j, such that: nums[i] + nums[j] = target
- Display the two indices separated by a space.
- Return the indices in any order.
- Display -1 -1 if no such pair exists.
- Use a nested-loop approach to solve the problem.
- Assume that at most one valid solution exists.

CODE

```
#include <stdio.h>


void two_sum(int *nums, int n, int target)
{
   for (int i = 0; i < n - 1; i++)
   {
      for (int j = i + 1; j < n; j++)
      {
         if (*(nums + i) + *(nums + j) == target)
         {
            printf("%d %d\n", i, j);
            return;
         }
      }
```

```
    }

    printf("-1 -1\n");
}

int main(void)
{
    int n, target;

    printf("Enter array size: ");
    scanf("%d", &n);

    if (n <= 0)
        return 1;

    int nums[n];

    printf("Enter array elements: ");
    for (int i = 0; i < n; i++)
        scanf("%d", nums + i);

    printf("Enter target: ");
    scanf("%d", &target);

    two_sum(nums, n, target);

    return 0;
}
```